

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: Voucher code generated with cryptographic checksum. Fail: Expired or duplicate voucher redeemed.	Core point conversion and exchange mechanism enabling interoperability across multiple merchant loyalty programs.
FR-002	Functional	The system shall calculate real-time dynamic conversion rates between partner merchant loyalty points and universal exchange credits based on merchant liquidity, point expiry, and market demand.	High	Pass: System returns a conversion quote guaranteed for 120 seconds with transparent fee spread. Fail: Quote generation times out (>2s) or uses stale rates (> 5m old).	Guarantees transparent, market-driven valuation while protecting platform liquidity against arbitrage.
FR-003	Functional	The system shall allow an Account Holder to link multiple merchant loyalty accounts and aggregate real-time reward point balances into a unified digital wallet interface.	Medium	Pass: Successfully authenticates and synchronizes live point balances across at least 3 distinct merchant APIs. Fail: Balance synchronization fails or leaks external auth credentials.	Consolidates fragmented reward point balances into a centralized dashboard for user convenience.
FR-004	Functional	The system shall enforce anti-fraud voucher locking by binding generated vouchers cryptographically to the Account Holder ID and requiring step-up MFA validation upon anomalous risk detection.	High	Pass: Voucher status transitions to 'Locked-Pending' and completes verification within 15 seconds. Fail: Voucher redeemed simultaneously on multiple channels or before identity verification.	Mitigates gift card fraud, double-spending, account takeovers, and automated bot scraping.
FR-005	Functional	The system shall issue verifiable digital gift cards featuring tamper-evident QR codes/barcodes and execute double-entry ledger settlement with the issuing Merchant Partner.	High	Pass: Activated gift card rendered within 500 ms and partner ledger reflects corresponding debit/credit settlement entry. Fail: Voucher issued with missing cryptographic hash or uncommitted ledger entry.	Completes end-to-end gift card redemption while maintaining auditable financial reconciliation with partner merchants.
NFR-001	Non-Functional (Performance & Security)	Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: Benchmarking tests confirm target latency (<250 ms) and security standards under simulated peak load (1,000 TPS). Fail: Operation latency exceeds 250 ms or ledger audit detects non-zero balance drift.	Critical for financial integrity, high-throughput consumer satisfaction, and maintaining zero accounting discrepancy across partner balances.
NFR-002	Non-Functional (Security & Compliance)	All sensitive credentials, voucher tokens, and transaction payloads must be encrypted in transit via TLS 1.3 and at rest via AES-256 with HSM-managed keys, adhering to PCI-DSS Level 1.	High	Pass: Vulnerability scan and static/dynamic cryptographic audit verify zero plaintext exposure of voucher keys in memory, logs, or storage. Fail: Plaintext secret or weak cipher detected during audit.	Safeguards high-value exchange assets against unauthorized exfiltration and ensures strict compliance with financial regulations.